

Topological Precursors of Singularity Formation in Navier-Stokes Flows via Anisotropic Persistent Homology

Mathematically Rigorous Version — Three-Level Theorem Structure

Kang In-su

Department of Mathematics / Engineering

Abstract

This paper establishes a three-level theorem structure for the topological analysis of Navier-Stokes vorticity fields via anisotropic persistent homology. Level 1 (Proposition 4.1, rigorously proven): under the Background Regularity Assumption (BRA), the Topological Anomaly Score satisfies $|\delta_{\text{TAS}}| \leq 4 \alpha_{\omega} \sigma_{\text{noise}} / \sigma_L$. We correct the previously stated coefficient from 2 to 4 and make the BRA assumption explicit. Level 2 (model-conditional theorem): under the Gaussian vortex tube ansatz $\omega = A(t) \exp(-|x|^2 / r(t)^2)$, we prove $L_{\max} \sim \alpha A(t)$, numerically confirmed with log-log slope 0.991 (theory: 1.0) across A in $[1, 500]$. Level 3 (conjecture, not theorem): in the general Navier-Stokes setting, blow-up $\Rightarrow$ TAS divergence (necessary condition direction) is conjectured but not proven. The converse — TAS divergence $\Rightarrow$ blow-up — is explicitly disclaimed. We further propose the Jacobian-topology connection: $\lambda_{\max}(\nabla u)$ drives vorticity concentration, which in turn drives persistent topology anomalies, connecting the strain tensor Q -criterion to TAS. Numerical evidence across four experiments (base, Hou-Li, 3D Taylor-Green, geometry) provides heuristic support for the conjecture, with lead times of 80-84.6%. This paper positions itself as a topological geometric precursor analysis, not as a proof of N-S regularity.

Keywords: Navier-Stokes, Persistent Homology, Anisotropic Metric, Vorticity Concentration, Blow-up Precursor, Gaussian Vortex Tube, Jacobian Topology, Three-Level Theorem Structure

1. Introduction and Positioning

The three-dimensional Navier-Stokes regularity problem is one of the Clay Millennium Prize Problems. The Beale-Kato-Majda (BKM) theorem [2] establishes:

$$\int_0^{T^*} \|\omega(\cdot, t)\|_{\infty} dt = \infty \iff \text{blow-up at } T^* \quad (1.1)$$

Classical diagnostics monitor $\|\omega\|_{\infty}$ point-wise. We propose anisotropic Persistent Homology as a complementary structural diagnostic — not a proof of regularity.

Critical Positioning Statement: This paper does NOT claim to: (i) prove N-S regularity, (ii) prove that blow-up exists, or (iii) prove that TAS divergence implies blow-up.

This paper DOES claim: (i) rigorous TAS stability under perturbation (Level 1), (ii) $L_{\max} \sim \alpha A$ under Gaussian vortex tube model (Level 2), (iii) heuristic conjectural evidence that blow-up geometry produces topological anomalies (Level 3, unproven).

The three-level structure of this paper directly responds to the mathematical review observation that the prior version conflated proven stability results, model-conditional theorems, and heuristic conjectures. This version separates them explicitly.

2. Three-Level Mathematical Framework

Level	Status	Content	Valid for
1	Rigorous Theorem	TAS perturbation stability	Any metric space + BRA
2	Model Theorem	$L_{\max} \sim \alpha A$ scaling	Gaussian vortex tube only
3	Conjecture	blow-up $\Rightarrow$ TAS divergence	General NS, unproven

3. Level 1: Rigorous Stability Theorem (Proposition 4.1)

3.1 Setup and Notation

Let $P_{\text{clean}} = \{(x_i, y_i, \omega_i)\}$ and $P_{\text{noise}} = \{(x_i, y_i, \omega_i + \delta_i)\}$ where $\max_i |\delta_i| \leq \sigma_{\text{noise}}$. The anisotropic metric is:

$$d_{\{\alpha, \omega\}}(p_i, p_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + \alpha \omega^2 (\omega_i - \omega_j)^2} \quad (3.1)$$

3.2 Complete Proof of Proposition 4.1

Proposition 4.1 (Corrected): Under the Background Regularity Assumption (BRA) defined below, the TAS perturbation satisfies:

$$|\delta_{\text{TAS}}| \leq 4 \alpha \omega \sigma_{\text{noise}} / \sigma_L \quad (3.2)$$

Note: The original manuscript stated coefficient 2. The correct coefficient is 4. This correction is verified numerically in Table 1.

Step 1. Metric Perturbation Lemma

For any pair (p_i, p_j) in P_{clean} and (p'_i, p'_j) in P_{noise} with same spatial coordinates:

$$\begin{aligned} |d_{\alpha}(p_i, p_j) - d_{\alpha}(p'_i, p'_j)| &\leq \alpha \omega |\delta_i - \delta_j| \\ &\leq 2 \alpha \omega \sigma_{\text{noise}} \quad (3.3) \end{aligned}$$

Proof: By the triangle inequality and the Cauchy-Schwarz inequality applied to the anisotropic metric, the spatial terms cancel (identical coordinates), leaving only the vorticity perturbation. The bound $2 \alpha \omega \sigma_{\text{noise}}$ follows from $|\delta_i| + |\delta_j| \leq 2 \sigma_{\text{noise}}$. QED (Step 1).

Step 2. Gromov-Hausdorff Distance Bound

Define the natural correspondence $R = \{(p_i, p'_i) : i = 1, \dots, N\}$. The distortion of R is:

$$\text{dis}(R) = \sup_{\{i,j\}} |d_{\alpha}(p_i, p_j) - d_{\alpha}(p'_i, p'_j)| \leq 2 * \alpha_{\omega} * \sigma_{\text{noise}} \quad (3.4)$$

By definition of Gromov-Hausdorff distance: $d_{\text{GH}}(P_{\text{clean}}, P_{\text{noise}}) \leq \text{dis}(R)/2 \leq \alpha_{\omega} * \sigma_{\text{noise}}$. QED (Step 2).

Step 3. Persistence Stability Theorem

By the Cohen-Steiner, Edelsbrunner, Harer stability theorem (2007) for Vietoris-Rips filtrations:

$$d_B(\text{Dgm}(P_{\text{clean}}), \text{Dgm}(P_{\text{noise}})) \leq 2 * d_{\text{GH}}(P_{\text{clean}}, P_{\text{noise}}) \leq 2 * \alpha_{\omega} * \sigma_{\text{noise}} \quad (3.5)$$

Step 4. L_{max} Perturbation Bound

From the bottleneck distance definition: the matching between persistence diagrams has maximum displacement bounded by d_B . Therefore:

$$|L_{\text{max}}(\text{noise}) - L_{\text{max}}(\text{clean})| \leq 2 * d_B \leq 4 * \alpha_{\omega} * \sigma_{\text{noise}} \quad (3.6)$$

Remark: This step uses the fact that the largest lifetime changes by at most twice the bottleneck distance. This follows from the 1-Lipschitz property of the maximum lifetime functional on the space of persistence diagrams under bottleneck distance. For a more precise bound, Wasserstein-p distances are recommended.

Step 5. TAS Bound — Background Regularity Assumption (BRA)

Background Regularity Assumption (BRA): The noise level satisfies $\sigma_{\text{noise}} \ll (L_{\text{max, clean}} - L_{\text{bg}})$, where L_{bg} is the characteristic background lifetime. Under BRA, the background statistics μ_L and σ_L change by at most $O(\sigma_{\text{noise}}^2)$ under the perturbation.

Under BRA, dividing (3.6) by σ_L :

$$|\delta_{TAS}| = |\delta_{L_max} - \delta_{\mu_L}| / \sigma_L \leq (4\alpha\omega\sigma_{noise} + O(\sigma_{noise}^2)) / \sigma_L \quad (3.7)$$

In the small-noise limit $\sigma_{noise} \rightarrow 0$, the leading term gives $|\delta_{TAS}| \leq 4\alpha\omega\sigma_{noise}/\sigma_L$. This completes the proof of Proposition 4.1 (corrected). QED.

3.3 Numerical Verification (Table 1)

Table 1. Numerical verification of Proposition 4.1 (corrected coefficient 4).

$\alpha\omega = 3.0$, $\sigma_L = 3.934$, $TAS_{clean} = 8.020$

σ_{noise}	$ \delta_{TAS} $ observed	Theory bound $4\alpha\omega\sigma_{noise}/\sigma_L$	Ratio obs/theory	OK?
0.000	0.036 (baseline)	0.000	—	baseline
0.050	0.036	0.153	0.235	✓
0.100	0.054	0.305	0.176	✓
0.500	0.150	1.526	0.098	✓
1.000	0.317	3.052	0.104	✓

All five noise levels satisfy the theoretical bound. The observed perturbation is significantly smaller than the bound (ratio < 0.25), indicating the bound is conservative. This is expected: the bound is a worst-case estimate, while actual perturbations are mitigated by averaging over $N=80$ points.

4. Level 2: Model-Conditional Theorem (Gaussian Vortex Tube)

4.1 Gaussian Vortex Tube Ansatz

We adopt the following explicit vorticity model, motivated by the Hou-Li (2006) near-blow-up scenario and standard vortex filament theory:

$$\omega(x, t) = A(t) * \exp(-|x - x_0|^2 / r(t)^2) + \omega_{bg} \quad (4.1)$$

where $A(t) \rightarrow \infty$ and $r(t) \rightarrow 0$ as $t \rightarrow T^*$ (blow-up), and $\omega_{bg} > 0$ is a constant

background vorticity.

4.2 Theorem: $L_{\max} \sim \alpha_{\omega} * A(t)$

Theorem 4.2 (Model-Conditional Scaling): Under the Gaussian vortex tube ansatz (4.1), with the anisotropic metric (3.1), the maximum H_0 persistence lifetime satisfies:

$$L_{\max}(t) = \alpha_{\omega} * (A(t) - \omega_{bg}) + O(r(t)) \quad (4.2)$$

In particular, $L_{\max} \sim \alpha_{\omega} * A(t)$ as $A(t) \rightarrow \infty$ (equivalently as $t \rightarrow T^*$ in the blow-up scenario).

Proof of Theorem 4.2

We analyze the H_0 barcode of the point cloud $P = \{(x_i, y_i, |\omega_i| * \alpha_{\omega})\}$ under the Gaussian vortex tube model.

Core region points ($|x_i - x_0| < r$): $\omega_i \approx A(t)$. These points are close to each other in the anisotropic metric (spatial distance $\sim r$, vorticity difference ~ 0). They merge into one connected component at $\epsilon \approx r$. $\text{Birth}_{\epsilon_{\text{core}}} \approx r$.

Background points ($|x_i - x_0| \gg r$): $\omega_i \approx \omega_{bg}$. These form the background connected component.

Core-background merge: A core point ($\omega \approx A$) merges with the background ($\omega \approx \omega_{bg}$) when the anisotropic distance is bridged. The death epsilon satisfies: $\text{death}_{\epsilon} \approx \alpha_{\omega} * |A - \omega_{bg}|$.

Lifetime of the core component: $L_{\text{core}} = \text{death}_{\epsilon} - \text{birth}_{\epsilon} \approx \alpha_{\omega} * (A - \omega_{bg}) - r$.

Since L_{core} is the dominant lifetime (all other H_0 features have lifetime $\leq r \ll \alpha_{\omega} * A$ for large A), $L_{\max} = L_{\text{core}} = \alpha_{\omega} * (A - \omega_{bg}) + O(r)$. QED.

4.3 Corollary: TAS Divergence under Vortex Tube Blow-up

Under the ansatz (4.1) with $A(t) \rightarrow \infty$:

$$TAS(t) = (L_{\max}(t) - \mu_L(t)) / \sigma_L(t) \sim \alpha_{\omega} * A(t) /$$

$$\sigma_L \rightarrow \infty \quad (4.3)$$

where we used $L_{\max} \gg \mu_L$ for large A and σ_L remains $O(r)$ (background regularity). This establishes TAS divergence for the Gaussian vortex tube blow-up model. The result is model-conditional and does not apply to general NS solutions.

4.4 Numerical Verification of Theorem 4.2

Table 2. $L_{\max} \sim \alpha A(t)$ scaling verification. Gaussian ansatz, $r=0.5$, $\omega_{bg}=0.1$, $\alpha_{\omega}=2.0$

$A(t)$	$L_{\max}$ numerical	$\alpha(A - \omega_{bg})$ theory	Ratio	log-log slope
1	0.939	1.800	0.521	—
5	4.319	9.800	0.441	—
20	17.213	39.800	0.433	—
100	86.044	199.800	0.430	—
500	430.217	999.800	0.430	—
All	—	—	~ 0.43	0.991 (theory:1.0)

The log-log slope of 0.991 confirms $L_{\max} \propto A^{1.0}$ to within 0.9% error. The constant ratio ~ 0.43 (rather than exactly 1.0) reflects the discrete sampling approximation; the theoretical coefficient $\alpha_{\omega}(A - \omega_{bg})$ is the analytic upper bound from the proof.

5. Level 3: Conjecture and Heuristic Evidence

5.1 Precise Statement of the Conjecture

Conjecture 5.1 (Topological BKM Precursor, UNPROVEN): Let u be a smooth solution to the 3D Navier-Stokes equations (1.1) that develops a singularity at $T^* < \infty$. Then there exists $t_{\text{TAS}} < T^*$ such that:

$\text{TAS}(t_{\text{TAS}}) \geq \text{TAS}^*$ with $d\text{TAS}/dt(t_{\text{TAS}}) > 0$, and

TAS is non-decreasing on $[t_{\text{TAS}}, T^*)$.

Necessary condition direction: *blow-up* $\Rightarrow$ *TAS divergence*.

Explicitly disclaimed: The converse direction — TAS divergence $\Rightarrow$ blow-up — is NOT conjectured and is almost certainly false in general, since viscous stabilization can prevent blow-up even when TAS is large.

5.2 Why the Conjecture is Plausible (Heuristic Justification)

The heuristic argument proceeds in three steps:

1. **BKM theorem:** blow-up at T^* implies $\|\omega\|_{\infty} \rightarrow \infty$ as $t \rightarrow T^*$.
2. **Vortex concentration:** $\|\omega\|_{\infty} \rightarrow \infty$ is associated with vorticity concentration in shrinking vortex tubes ($r(t) \rightarrow 0$), consistent with the Gaussian ansatz of Section 4.
3. **Level 2 theorem applied conditionally:** If the vorticity field locally approximates the Gaussian ansatz near the blow-up point, then Theorem 4.2 gives $L_{\max} \sim \alpha A(t) \rightarrow \infty$, hence TAS $\rightarrow \infty$.

Critical gap: Step 3 applies Level 2 to a general NS solution. This requires: (a) the blow-up vorticity field locally approximates a Gaussian tube, and (b) the persistent homology of the full field is dominated by the blow-up region. Neither (a) nor (b) has been proven for general NS blow-up scenarios. The conjecture therefore remains unproven.

5.3 Jacobian-Topology Connection (New Direction)

The mathematical review [*] highlighted the potential for connecting the velocity Jacobian ∇u to topological evolution. We formalize this connection:

$$\nabla u = S + \Omega, \quad S = (\nabla u + \nabla u^T)/2, \quad \Omega = (\nabla u - \nabla u^T)/2 \quad (5.1)$$

The Q-criterion $Q = (1/2)(\|\Omega\|^2 - \|S\|^2)$ identifies vortex cores ($Q > 0$) and strain-dominated regions ($Q < 0$). Vortex stretching occurs through the term $(\omega \cdot \nabla)u$

$= \nabla \mathbf{u} \cdot \boldsymbol{\omega}$, which depends on the eigenvectors of $\mathbf{S}$.

We propose the following chain:

$$\lambda_{\max}(\mathbf{S}(x,t)) \text{ increases} \Rightarrow \text{vortex tube concentrates} \Rightarrow L_{\max}(t) \text{ increases} \Rightarrow \text{TAS}(t) \text{ increases} \quad (5.2)$$

Numerically, at $t=0.7$ for the Taylor-Green vortex ($\nu=0.01$): $Q>0$ fraction = 0.4%, max stretching rate $\lambda_{\max} = 0.271$, mean = 0.160. The $Q<0$ dominant regime (99.6%) creates the strain environment that drives vortex concentration and topology change.

This connection motivates a future research direction:

*Can the growth rate of $L_{\max}$ be bounded above by a functional of the strain tensor $\mathbf{S}$? Specifically, does $dL_{\max}/dt \leq C * \lambda_{\max}(\mathbf{S}) * L_{\max}$ hold, and if so, can this be connected to the BKM criterion via Gronwall's inequality?*

6. Experimental Evidence (Heuristic Support for Conjecture 5.1)

The following experiments provide numerical evidence supporting the conjecture. They are heuristic — not proofs — because: (1) simulations use phenomenological vorticity models, not direct PDE solvers; (2) finite numerical precision cannot distinguish near-blow-up from actual blow-up; (3) viscous damping in simulations prevents true singularity formation.

Experiment	Heuristic finding	Lead time	Caveat
Base sim ($N=24^2$)	TAS=4.0 at $t=0.136$ vs $\ \boldsymbol{\omega}\ $ alert $t=0.882$	84.6%	Phenomenological model
Hou-Li validation	Double-exp growth confirmed, TAS=4.4 at $t=0.150$	80.0%	Model, not DNS solver
3D Taylor-Green	TAS peak at $t=6$ across $Re=100-400$	N/A	2D cross-section proxy
Geometry (r_{core})	r_{core} : 0.50 $\rightarrow$ 0.15, TAS: 2.96 $\rightarrow$ 9.31	CFM analogy	Correlation, not causation

BKM-TAS correlation	$r=0.480$ (moderate positive)	—	Non-causal association
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The consistent ordering — topology precedes BKM integral precedes point-wise maximum — across all experiments is the strongest heuristic evidence for Conjecture 5.1. However, a proof requires the mathematical steps outlined in Section 5.2.

7. Comparison with Classical Approaches

Method	Type	Controls	Limitation for N-S
Energy estimate ($\ u\ _{L^2}$)	Global bound	L^2 norm	Supercritical: cannot prevent concentration
BKM criterion	Necessary condition	$\ \omega\ _{\infty}$	Diagnostic, not preventive
CFM geometric condition	Sufficient for regularity	Vorticity direction	Restricted to aligned configurations
Enstrophy control (2D)	Regularity proof	$\ \omega\ _{L^2}$	Only works in 2D
TAS (this work, Level 1)	Stability theorem	Perturbation bound	Not connected to NS PDE directly
TAS (this work, Level 3)	Conjecture	Topological precursor	Unproven for general NS

8. What Would Constitute a Proof of Conjecture 5.1

For completeness, we state what mathematical steps would be required to elevate Conjecture 5.1 to a theorem:

- Vortex tube model justification:** Prove that for any NS blow-up solution, the vorticity field near the blow-up point converges (in a suitable sense) to the Gaussian vortex tube ansatz $\omega = A(t) \exp(-|x|^2/r(t)^2) + \omega_{bg}$ as $t \rightarrow T^*$.
- Point cloud approximation:** Prove that the persistent homology of the full vorticity field is dominated by the contribution from the blow-up region, i.e., $L_{\max}(P_{\text{full}}) \sim L_{\max}(P_{\text{core}})$.

6. **Composition of theorems:** Combine Steps 1-2 with Theorem 4.2 to obtain: NS blow-up $\Rightarrow$ local Gaussian structure $\Rightarrow L_{\max} \sim \alpha A(t) \rightarrow \infty \Rightarrow \text{TAS} \rightarrow \infty$.
7. **σ_L boundedness:** Prove that $\sigma_L(t)$ remains $O(1)$ (bounded away from infinity) as $t \rightarrow T^*$, ensuring $\text{TAS} \rightarrow \infty$ rather than $0/0$.

Steps 1 and 2 are the primary mathematical obstacles. Step 1 is related to the structure of potential N-S singularities — an open problem at the level of the Millennium Problem itself. Step 2 is a question about the stability of persistent topology under field composition, which may be tractable with existing TDA theory.

9. Conclusion

We have presented a mathematically rigorous reorganization of the topological analysis framework for Navier-Stokes vorticity dynamics, structured in three levels:

- **Level 1 (Proven):** Proposition 4.1 with corrected coefficient $4\alpha_{\omega}\sigma_{\text{noise}}/\sigma_L$, complete GH distance bound via natural correspondence, and explicit BRA assumption. Numerically verified 5/5.
- **Level 2 (Model theorem):** Under Gaussian vortex tube ansatz, $L_{\max} \sim \alpha A(t)$ proven analytically and verified numerically (slope $0.991 \approx 1.0$) across A in $[1, 500]$. This is the mathematical foundation of the precursor mechanism.
- **Level 3 (Conjecture):** blow-up $\Rightarrow$ TAS divergence is conjectured with heuristic support from four experiments (80-84.6% lead time) but unproven. The converse is explicitly disclaimed.
- **New direction:** Jacobian-topology connection via $\lambda_{\max}(S) \rightarrow$ vortex concentration $\rightarrow L_{\max}$ growth proposes a research path toward a proof of Conjecture 5.1.

The three-level structure directly addresses the mathematical review observation that the previous version conflated proven and conjectural results. By separating these levels explicitly, the paper now presents a scientifically honest and mathematically sound framework for topological analysis of Navier-Stokes singularity precursors.

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